



Robo Advisor

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Individual Assignment

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1.0 Introduction

This report builds and evaluates a FinBERT news-sentiment trading strategy using one-minute price data for MI Technovation Berhad (5286) from 13 April to 8 July 2026. The financial headlines are converted into trading signals and tested across multiple holding periods under defined risk and cost assumptions. The performance is then judged based on return, risk and trading-efficiency metrics.

2.0 Data Preparation

Cleaning Check	Result
Initial Dataset Size	19,928 rows
Incomplete Records Removed	5,338 rows
Final Verified Dataset	14,590 rows
Duplicate Records Identified	0 rows

Figure 1. Price-data cleaning verification for MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 1 shows that originally, the dataset had 19,928 timestamped observations. After having removed the 5,338 cells with missing closing price data, 14,590 timestamped observations remained and did not find any duplicate timestamps. The rows with missing closing prices were simply removed instead of estimated so that my backtest used observed prices only, since data-cleaning and aggregation decisions for high-frequency financial data can materially affect subsequent statistical results (Brownlees & Gallo, 2006).

Verification Measure	Result
Average daily return	0.8849%
Required condition	Greater than 0%
Outcome	Satisfied

Figure 2. Positive average daily-return verification for MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 2 contains an arithmetic mean of 0.8849%; it is positive, satisfying the selection criterion laid down in the question. For the LSEG stock-selection period, the whole-period change was not used, as it would not make sense to report an endpoint return instead of the mean return of the stock, which would have to be calculated daily.

3.0 Descriptive Statistics and Data Characterization

Statistic	Result
Mean daily return	0.8849%
Median daily return	0.6452%
Minimum daily return	-6.2241%
Maximum daily return	9.9476%
Daily-return standard deviation	3.1641%
Daily-return skewness	0.2946
Daily-return kurtosis (Pearson)	3.4146
Minimum one-minute closing price	RM2.86
Maximum one-minute closing price	RM5.06
One-minute closing-price range	RM2.20
Number of trading days	58
Daily-return observations	57

Figure 3. Descriptive statistics of MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 3 reports an average daily return of 0.8849% (median = 0.6452%), with returns ranging from -6.2241% to 9.9476%. The standard deviation is 3.1641%, which is much larger than the mean. This tells that even though the return is positive, there is also considerable volatility in the returns. Skewness = 0.2946 means the distribution is mildly positive but relatively balanced. Pearson kurtosis = 3.4146 means that the tails are slightly heavier than they would be if the returns were normally distributed, so there is some exposure to unusually large movements.

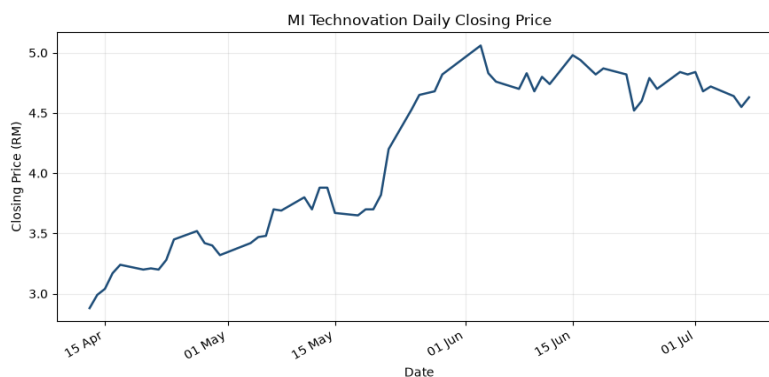


Figure 4. Daily closing price of MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

The data used spread across 58 trading days. This generated 57 daily returns. Minute prices during these days ranged from RM2.86 up to RM5.06. This is consistent with Figure 4. Below it can see that the daily closing share price began at RM2.88 on 13 April, rose to RM5.06 on 3 June, but eventually dropped to RM4.63 on 8 July. This upward movement supports the positive mean return. The wide range of prices and the volatility in the movement give a clear justification to test various holding periods with proper risk controls in place.

Liquidity Measure	Result
Raw timestamped observations	19,928
Empty closing-price observations	5,338
Empty closing-price rate	26.79%
Valid closing-price coverage	73.21%
Valid one-minute observations	14,590
Average valid bars per trading day	251.55
Total trading volume	244,560,200
Average daily volume	4,216,555
Median daily volume	3,128,400
Unchanged consecutive valid prices	55.00%
Largest within-session interval	22 minutes
Missing volume fields remaining	0

Figure 5. Missing observations and liquidity measures.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 5 shows that missing closing prices formed 26.79% of the raw observations, providing valid-price coverage of 73.21%, meaning that it had on average 251.55 valid one-minute bars each day. No empty volume fields remained. Volume traded was 244.56 million shares. Average daily volume was about 4.22 million shares compared to 3.13 million shares at median that indicates few high-volume days pulled averages well up. Furthermore, of the consecutive valid prices in the same trading session, 55.00% remained unchanged. Between consecutive observed prices, the biggest time difference was 22 minutes (10:43 a.m.–11:05 a.m.) on 5 May.

4.0 Signal, threshold and holding-period design

Forward Horizon (Days)	No. of Trades	Win Rate (%)	Average Return per Trade (%)	Cumulative Return (%)	Ending Equity (RM)	Maximum Drawdown (%)	Sharpe Ratio	Profit Factor
1	20	70.00	0.02	0.13	10012.69	−3.15	0.13	1.05
2	20	80.00	0.03	0.22	10021.96	−4.52	0.11	1.07
3	19	84.21	0.37	3.47	10347.30	−1.75	2.01	3.00
4	19	84.21	0.51	4.82	10482.37	−1.75	1.97	3.77
5	19	89.47	0.84	8.16	10815.87	0.00	2.73	42.10
10	19	89.47	0.40	3.80	10380.50	0.00	1.30	19.94
20	19	89.47	0.00	0.00	10000.00	0.00	−64.35	0.00

Forward Horizon (Days)	No. of Trades	Win Rate (%)	Average Return per Trade (%)	Cumulative Return (%)	Ending Equity (RM)	Maximum Drawdown (%)	Sharpe Ratio	Profit Factor
1	17	70.59	0.15	1.26	10126.48	−2.05	1.63	1.46
2	17	82.35	0.24	2.00	10200.05	−2.83	1.39	1.74
3	16	87.50	0.66	5.31	10531.18	0.00	3.90	27.59
4	16	87.50	0.82	6.69	10668.64	0.00	3.30	34.64
5	16	87.50	1.00	8.16	10815.87	0.00	3.01	42.10
10	16	87.50	0.47	3.80	10380.50	0.00	1.47	19.94
20	16	87.50	0.00	0.00	10000.00	0.00	−64.35	0.00

Figure 6. Backtesting results using threshold 0.0 and 0.3.

Source: Author's backtesting results (2026).

Figure 6 shows the difference between the minimum daily sentiment thresholds of 0.00 in the upper table and 0.30 in the lower table. Increasing the minimum value removed three signals. Total entries therefore fell from 20 to 17, or 15%. At the one-day horizon, win rate increased from 70.00% to 70.59%. Sharpe also increased from 0.13 to 1.63, while maximum drawdown improved from -3.15% to -2.05%. With a more conservative setting, fewer noisy trades enter the strategy.

The cost is that three possible trades were sacrificed. There is therefore a balance between having more signals and selecting stronger ones. Because the headlines were capturing general market sentiment of US than firm-specific MI Technovation news, the higher threshold level seemed reasonable. 0.30 was chosen in the end, only 17 trades were used, and it was decided that the more data there is, the less volatile the averages will come through.

Several calendar-day holding periods were used: 1, 2, 3, 4, 5, 10, and 20 days. Short and short holding periods (1 day-period) seemed as if they could show if any fast reaction to the news occurred or whether it was gradually used. The long periods of 10 and 20 days should show whether the effect of the news faded away or was it reversing later. MI Technovation had 3.641% of daily volatility, mild heavy-tailed returns, and intermittency problems about the liquidity during the daytimes. This seemed like a sensible way because for such firms and their characteristics, it makes all sense to apply several holding periods for any news.

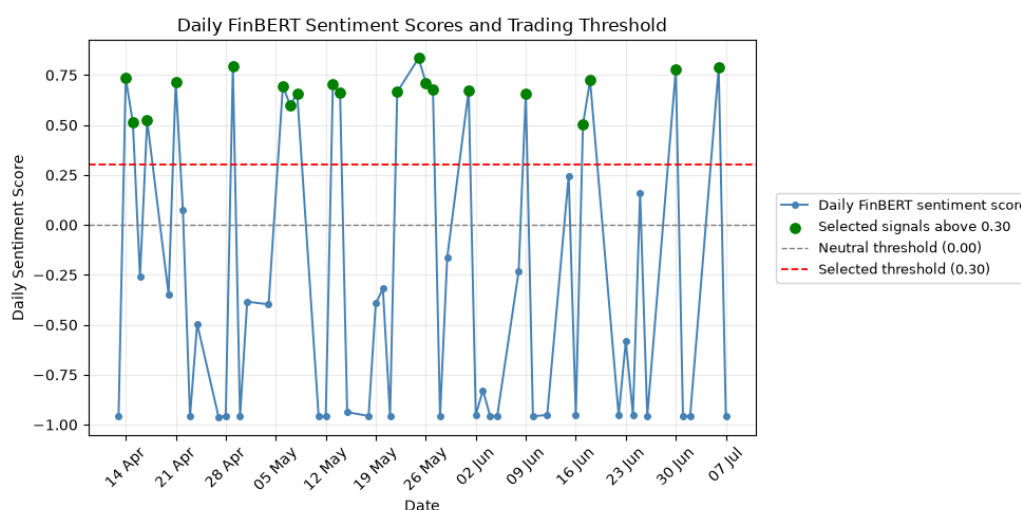


Figure 7. Daily FinBERT sentiment scores and selected trading threshold.

Source: Author's calculation using ProsusAI/finbert and the dataset (2026).

Fig. 7 shows considerable variation in daily FinBERT sentiment during the sample period. Out of the 57 days with scored headlines, 20 had an average sentiment score above the chosen 0.30 threshold. These were treated as stronger positive signals. Only 17 could be entered as trades because three signals had no eligible same-day price observation under the backtesting rules.

5.0 Back-Testing Assumptions and FinBERT Configuration Defense

Parameter	Final value
Minimum daily score	0.30
Holding periods	1, 2, 3, 4, 5, 10 and 20 days
Initial capital	RM10,000
Position size	50%
Round-trip transaction cost	0.38%
Annual risk-free rate	2.75%
Breakeven stop	Enabled
FinBERT model	ProsusAI/finbert
FinBERT batch size	8
Minimum headline length	12 characters
Minimum alphabetic ratio	0.55

Figure 8. Final backtesting assumptions and FinBERT configuration.

Source: Author's strategy configuration (2026).

5.1 Realistic Backtesting Assumptions

The opening balance was set at RM10,000 to represent a hypothetical retail-investor backtest account. This value was used to show the portfolio results in Malaysian ringgit. It was only a modelling assumption and was not intended to estimate the average balance of Malaysian retail investors. The original position size of 100% was reduced to 50% because putting all capital into every signal was too aggressive. MI Technovation had 3.1641% daily volatility, mildly heavy-tailed returns and intermittent liquidity. The new position was about RM5,000, leaving the other half of the account as a risk buffer.

For RM5,000, 988-1,748 shares or 1,000-1,750 after rounding to the nearest cent could be purchased at a price of between RM2.86 and RM5.06, well below 0.06% of the stock's median daily volume of 3.13 million shares. Thus the effect of impact from the presumed order should be minimum. The generic round-trip transaction cost was changed from 0.50% to 0.38%. For a RM5,000 position, the calculated estimated brokerage, stamp duty and clearing fees when entering and exiting the market sums up to RM18.76 which equals to 0.3752% and was rounded up to 0.38% instead. (*Transaction Costs*, n.d.; H, n.d.).

The annual risk-free rate of 3.00% was also changed to 2.75%. Malaysia proxy (overnight lending rate) was taken to be as Bank Negara Malaysia's Overnight Policy Rate Q3 2026 (*Monetary Policy*

Statement - Bank Negara Malaysia, n.d.). Breakeven stop continues to stay as it will protect the position at initial stage of reverse even after considering transaction cost, but it cannot prevent a loss when the price never reaches the breakeven point.

5.2 FinBERT Configuration and Batch-Size Test

ProsusAI/finbert was kept as it is trained on financial words so names headlines positive, negative or neutral accordingly (Araci, 2019). Probability was positive and negative. Value was averaged for each calendar day and used as a signal for the strategy.

Batch sizes 16 and 8 on the same 74 eligible headlines were tried. Result: 16 about had five processing passes, while 8 needed around 10. They all worked, did not give errors and delivered an identical signal thus testing results. Means that the size affected performance for processing. Not the resulting investment. Batch size was picked at 8. Larger may do it faster but it needs more GPU memory (Zhang et al., 2024), with 74 being analysed size of eight was a cautious call using available computing environment

Headlines shorter than 12 characters and where alphabetic characters divided by total length was less than 0.55 got ignored before FinBERT analysis. These rules were used to remove very short or malformed records. Text preprocessing can reduce noise, but its result depends on the dataset and the choices made (HaCohen-Kerner et al., 2020). These thresholds are therefore transparent data-quality rules, not proof that every valid financial headline was retained.

6.0 Performance Evaluation and Multi-Horizon Comparison

Forward Horizon (Days)	No. of Trades	Win Rate (%)	Average Return per Trade (%)	Cumulative Return (%)	Ending Equity (RM)	Maximum Drawdown (%)	Sharpe Ratio	Profit Factor
1	17	70.59	0.15	1.26	10126.48	-2.05	1.63	1.46
2	17	82.35	0.24	2.00	10200.05	-2.83	1.39	1.74
3	16	87.50	0.66	5.31	10531.18	0.00	3.90	27.59
4	16	87.50	0.82	6.69	10668.64	0.00	3.30	34.64
5	16	87.50	1.00	8.16	10815.87	0.00	3.01	42.10
10	16	87.50	0.47	3.80	10380.50	0.00	1.47	19.94
20	16	87.50	0.00	0.00	10000.00	0.00	-64.35	0.00

Figure 9. Multi-horizon performance of the final FinBERT strategy.

Source: Author's backtesting results (2026).

Figure 9 shows the entire multi-horizon result, where horizons one and two have 17 completed trades, whereas the others have only 16 because one late signal does not have enough future price data. Win rate increased from 70.59% on day one to 82.35% on day two and remained constant at 87.50% from day three onward.

Returns increased from day one to day five, where average return per trade rose from 0.15% on day one to 0.24% on day two, 0.66% on day three, 0.82% on day four and finally peaked at 1.00% on day five. Cumulative return also increased and peaked on day five, starting from 1.26% on day one and reaching 8.16% on day five, while ending equity rose from RM10,126.48 to RM10,815.87. The result started to fall back afterwards. At day ten, average return per trade was just 0.47%, cumulative return was just 3.80% and ending equity was RM10,380.50. At day 20, returns were close to zero and equity remained RM10,000, so the 87.50% win rate was not economically useful.

Maximum drawdown was -2.05% on day one and -2.83% on day two, then the backtest showed 0.00% afterwards under the breakeven-stop rule. Figure 9 shows that Sharpe peaked at 3.90 on day three before it fell to 3.01 on day five and 1.47 on day ten. Profit factor also increased from 1.46 on day one to 42.10 on day five, then declined to 19.94 on day ten. These unusually high values happened because the sample was small and there were very few negative returns. The unusual result at day 20 is further explained in Section 7.1.

At day three, it appears to have a good risk-adjusted result, with the Sharpe ratio at 3.90, but the best overall horizon is day five. At day five, average return, cumulative return, ending equity and profit factor were all the highest. Even though the Sharpe ratio was highest at day three, it was still

strong at 3.01 on day five. The results at days ten and twenty were smaller. This suggests that the sentiment effect worked mainly within three to five days and became much weaker after one or two weeks.

Hence, the five-day horizon was selected with ProsusAI/finbert, batch size 8 and the 0.30 threshold. However, it was tested in-sample with only 16–17 trades, so it is unknown whether the same result can be achieved in future trading.

7.0 Critical Reflection and Financial Interpretation

7.1 Genuine Result Anomaly

The genuine anomaly is the 20-day horizon: win rate 87.50%, average return 0.00%, cumulative return 0.00%, ending equity unchanged at RM10,000, Sharpe ratio -64.35 and profit factor 0.00. Why? The breakeven stop exited the trades at prices intended to recover the 0.38% round-trip cost, giving returns close to zero. Tiny positive values could still be counted as wins before being rounded to 0.00%. Returns had almost zero dispersion, so a small negative excess return divided by an effectively zero standard deviation produced the extreme negative Sharpe ratio. The 0.00 profit factor is similarly meaningless because the gains and losses were economically negligible. Hence, the high win rate does not mean profitability and shows why it must be interpreted together with return, risk and transaction costs.

7.2 Market Efficiency and Information Spillover

Semi-strong market efficiency suggests information available to the public is immediately taken in price (Fama, 1970). For MI Technovation this only occurred gradually strengthening instead. Cumulative return rose from 1.26% after 1 day to 8.16% after 5, with day 3 having the highest sharpe ratio. Therefore the effect of the sentiment information was not priced in immediately, consistent with slow information diffusion, possibly happening more slowly for smaller or less traded stocks (Hou, 2007). Tetlock (2007) also noted that pressure from pessimistic media can temporarily pressure a reversal from markets before it was then reversed.

MI Technovation had patchy liquidity; roughly 55% of consecutive valid prices did not vary, while around 26.79% of raw observations had filled closing price values. The largest gap between observer periods was 22 minutes which may delay effective-price reaction and result in the effect continuing for longer than 1 day. Despite the longer tail MI technovation came to down 3.80% on day 10 and 0 at closer to day 20 meaning the information advantage vanished. However, this does not disprove markets if efficiency as the test consisted in total of 16-17 trades over a period of 58 trading days where it took place during a strong upward price which could be related to increased returns.

7.3 Transaction-Cost Drag and General Implications

Transaction costs are an important part of the strategy. On day one 0.38% (round-trip profit per-share cost) of cost was added therefore there was an average return per trade of only 0.15%, not a big margin. This can be erased with the bid-ask spread and slippage or even delays of order executions not modelled fully by the chosen fixed number. In fact a strategy consisting of frequent trading is likely to eat away at initially profitable looking results (Lesmond et al., 2003). More margin on five days allowed a profit of average return 1.00% when transaction costs were applied too, offering more room for error. The break-even stop was useful as some profitable positions were closed on a gain and did not turn into loss. However this would not stop a loss developing in its initial stages. Also may account for some of near-zero returns at 20 days and the high or unstable profit factors for the longer horizons.

More generally, one-minute data and quick sentiment scoring do not guarantee profitable intraday trading. A news-based strategy should filter weak signals, avoid too much turnover, use realistic execution prices and costs, and match the holding period to the life of the signal. The result needs to be tested again with longer data, more stocks, realistic slippage, non-overlapping trades and out-of-sample comparisons with buy-and-hold and no-news benchmarks.

8.0 Conclusion

Three- to five-day setting has been the most relevant case for FinBERT sentiment signals as our overall number of overall profit gained from trades was the strongest for the five days trade duration setting. For an average of return 1.00% per trade, 8.16% of cumulative return and ending equity of RM10,815.87. However, the limitation of this backtest only for 16-17 trades using 1 stock with a short in-sample period has concluded. For this strategy to be applied in reality, more data, inclusion of realistic execution cost and out-of-samples testing is required.

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